2005 年真题参考答案

一、填空题

(1)2. (2)$xy = 2$. (3)$2\mathrm{e}\mathrm{d}x + (\mathrm{e} + 2)\mathrm{d}y$. (4)$\frac{1}{2}$. (5)$\frac{13}{48}$. (6)0.4; 0.1.

二、选择题

(7) B. (8) A. (9) D. (10) B. (11) C. (12) A. (13) D. (14) C.

三、解答题

(15)$\frac{3}{2}$.

(16)$x^2\frac{\partial^2 g}{\partial x^2} - y^2\frac{\partial^2 g}{\partial y^2} = \frac{2y}{x}f'\left(\frac{y}{x}\right)$.

(17)$\frac{\pi}{4} - \frac{1}{3}$.

(18)$S(x) = \begin{cases} \frac{1}{2x}\ln\frac{1+x}{1-x} - \frac{1}{1-x^2}, & x \in (-1,1) \setminus \{0\}, \\ 0, & x = 0. \end{cases}$

(19) 证明略.(可考虑$F(x) = \int_0^x g(t)f'(t)\mathrm{d}t + \int_0^1 f(t)g'(t)\mathrm{d}t - f(x)g(1)$, $x \in [0,1]$,证明$F'(x) \leqslant 0$.)

(20)$a = 2$, $b = 1$, $c = 2$.

(21)(Ⅰ)$\boldsymbol{P}^{\mathrm{T}}\boldsymbol{D}\boldsymbol{P} = \begin{pmatrix} \boldsymbol{A} & \boldsymbol{O} \\ \boldsymbol{O} & \boldsymbol{B} - \boldsymbol{C}^{\mathrm{T}}\boldsymbol{A}^{-1}\boldsymbol{C} \end{pmatrix}$.

(Ⅱ)矩阵$\boldsymbol{B} - \boldsymbol{C}^{\mathrm{T}}\boldsymbol{A}^{-1}\boldsymbol{C}$是正定矩阵.证明略.($\begin{pmatrix} \boldsymbol{A} & \boldsymbol{O} \\ \boldsymbol{O} & \boldsymbol{B} - \boldsymbol{C}^{\mathrm{T}}\boldsymbol{A}^{-1}\boldsymbol{C} \end{pmatrix}$是正定矩阵,其各阶顺序主子式均大于零,证明$\boldsymbol{B} - \boldsymbol{C}^{\mathrm{T}}\boldsymbol{A}^{-1}\boldsymbol{C}$的各阶顺序主子式均大于零.)

(22)(Ⅰ)$f_X(x) = \begin{cases} 2x, & 0 < x < 1, \\ 0, & \text{其他}, \end{cases}$ $f_Y(y) = \begin{cases} 1 - \frac{y}{2}, & 0 < y < 2, \\ 0, & \text{其他}. \end{cases}$

(Ⅱ)$f_Z(z) = \begin{cases} 1 - \frac{z}{2}, & 0 < z < 2, \\ 0, & \text{其他}. \end{cases}$

(Ⅲ)$\frac{3}{4}$.

(23)(Ⅰ)$D(Y_i) = \frac{n-1}{n}\sigma^2$, $i = 1, 2, \cdots, n$. (Ⅱ)$\mathrm{Cov}(Y_1, Y_n) = -\frac{1}{n}\sigma^2$. (Ⅲ)$c = \frac{n}{2(n-2)}$.